

Supervisor Research Playbook — v3 (Final)

CSS Final Paper · Expertise reproduction on Stack Overflow after generative AI

Leaner and causally honest. Scoped to what one MSc student can defend in one month.

v3 corrects v2 and removes complexity. Same gates, four tracks, one-month shape, literature-first, writing-in-parallel. Where a step doesn't move a rubric criterion, it's gone.

Read first — the scope you're actually committing to

- The **robust, defensible target**: a descriptive study of the **conversion function** (unconditional) + **one clean event-study difference-in-differences** on a validated binary treatment, on **one** Stack Exchange site (or a defensible tag-subset).
 - The **MVP (descriptive-only)** is a **complete paper**. The DiD is the *upgrade*, not the foundation. If it fails, you still submit a strong paper.
 - The full causal apparatus is a stretch for one student in a month. This plan protects you by locking the causal core early and de-scoping the rest.
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Fix #1 (the one that matters most): the estimand

v2's headline — the conversion function *conditional on persisting* — **conditions on a post-treatment collider**. ChatGPT changes *who persists* (weaker newcomers exit), so survivors are selected by the treatment; comparing them across cohorts conflates a real pipeline change with a change in who counts as a survivor. A DiD reviewer rejects this on sight.

- **Primary estimand — UNCONDITIONAL on persistence**: of all newcomers who **join** in cohort C, the share that enter the answerer pipeline within a fixed window; and time-to-first-accepted-answer. Attrition folds into the outcome — which is the honest meaning of "does the intake still produce answerers." No collider.
- **Secondary, explicitly caveated**: "conditional on persisting, do survivors mature slower?" Never the headline.

Fix #2: construct honesty

- Real SO expertise takes years; a 12-month window cannot observe it. **Reframe the endpoint to the *first rungs of the answerer pipeline*** — first accepted answer, transition to net-positive answering. You measure *entry into expertise formation*, not full expertise. Say so; it's still novel.
- You can only speak to reproduction **on Stack Overflow**, not to whether developers become experts elsewhere (via AI/Discord). **Scope the title and claims to the**

platform. External validity is a stated limit.

Fix #3: identification (one primary spec)

- **The DiD is the identification.** Cohort trajectories are descriptive context — not a co-equal design. The pre/post cohort arm has no counterfactual.
- **Primary specification: a single event-study** — cohort conversion rate over time, high- vs low-substitutability tags — which gives the pre-trend check *and* the effect in one figure. Drop the separate 2×2 and standalone cohort-comparison as primary.
- **Unit of analysis = tag (or tag-period), decided *jointly* with treatment assignment.** Substitutability is a tag property; users span tags, so user-level treatment is ill-defined and entangled with the AI-driven shift in who-asks-what. Lock unit + assignment together in **Phase 4**.
- **One primary outcome.** Reputation decomposition, accepted-answer counts, and tag breadth become labeled *exploratory*.

Fix #4: treatment validity (temporal)

Validating "which tags were AI-substitutable" with *today's* model is anachronistic (the treatment period is GPT-3.5-era). **Prefer a contemporaneous, behavioral signal of *realized* substitutability** — the magnitude of the on-platform question-volume drop per tag after Nov 2022, plus public training-data volume per tag — and use a period-appropriate model if you run an LLM-answer check. **GATE T** tests this; failure → drop the DiD.

Continuous tracks (four — unchanged)

- **R — References:** Zotero; cite only what you've opened; one style. (*Fabrication = rejection.*)
- **C — Reproducibility:** git, seeds, pinned env, **recorded dump snapshot**; ship sample + access + snapshot ID.
- **W — Write as you go:** Week-1 contribution statement + paper skeleton + rough Intro; Methods at freeze; Results as figures land.
- **D — Decision & Assumption Log:** every contestable choice + assumption, with rationale + date. Becomes Methods/Limitations.

Phase structure (compact)

Tags: [E] exploratory · [C] committing · [CP] checkpoint.

#	Phase	Do	Gate/output
0	Orient ☒ [E]	Internalise rubric	—
1	Map literature ☒ [E]	Cluster the field	—
2	Literature matrix [E]	6 core papers → matrix (incl. "what they did NOT do")	Matrix
3	Validate gap [E]	Confirm conversion-function gap; write contribution statement	GATE A
4	Lock the causal core [C] (<i>moved up</i>)	Lock estimand (unconditional) , unit = tag , treatment-assignment , primary outcome , window ; <i>propose</i> remaining ops	Locked core one-pager
5	Feasibility + power + split [E]	Computability + joins + censoring; count post-cohort N × base rate (power) ; declare develop/confirm split; no real pre/post peeking	GATE B (kill)
6	Build & validate treatment [E→C]	Per-tag substitutability from contemporaneous signals; confound checks (tag age/popularity/difficulty)	GATE T
7	Instructor meeting [CP]	Bring the brief (below)	GATE C
8	Freeze + pre-analysis plan [C]	Lock estimator (event-study), clustering (tag), composition control; timestamped PAP to git	PAP committed
9	Build pipeline [C]	Raw → cohort panel; record snapshot, pin env	Clean panel
10	Exploratory (develop set) [E]	Descriptives; secular trend; substitutability gradient — confirm set sealed	Exploratory figs
11	Parallel-trends FIRST + confirmatory (confirm set) [C]	Event-study pre-trend must pass; then frozen spec + robustness	GATE D
12	Interpret [C]	Link to LPP + Burtch; statistical-vs-substantive; "consistent with, not proof of"	—
13	Write (consolidate) [C]	Intro/Discussion/Limits/Abstract; Week-4 lit re-scan	Full draft
14	Audit + package + submit [C][CP opt]	Verify every ref; clean-run repro; confirm registration	Submission

v2's fuller 9-field notes still apply where unchanged.

Decision gates

- **A (Ph.3):** Gap survives + distinct-in-kind from Burtch → proceed / revise.
- **B (Ph.5) kill-gate:** Variables computable **AND post-cohort $N \times$ base rate gives minimal power AND** split declared → proceed / pivot / fall back. (*New power criterion — the Week-2 wall, caught in Week 1.*)
- **T (Ph.6):** Substitutability proxy tracks realized substitutability + survives confound checks → proceed / repair / **drop DiD → cohort-comparison MVP.**
- **C (Ph.7):** Instructor go / adjust / reframe.
- **D (Ph.11):** Event-study pre-period flat → confirmatory DiD / **invoke MVP.**

Contingency & fallbacks

- **MVP:** unconditional conversion-function **descriptive** cohort study — a complete, rubric-accepted paper without the DiD. **Pre-choose the smaller-scope site/tag-subset today.**
- **(a) Pipeline overruns** → smaller site / tag-subset / SEDE.
- **(b) Parallel trends fail (GATE D)** → drop causal DiD; present cohort + event-study descriptively (= MVP).
- **(c) Treatment weak (GATE T)** → drop DiD; unconditional cohort comparison stands alone.

Degrade only as far as necessary; the DiD is the ambition, not the floor.

Instructor meeting brief — bring this one page

- **RQ:** *After ChatGPT, does a smaller share of Stack Overflow newcomer cohorts enter the answerer pipeline within their first year — and is the decline concentrated in AI-substitutable topics?*
- **Contribution:** *First measure of the unconditional conversion function (intake → answerer), not activity or inflow; operationalises legitimate peripheral participation on a knowledge commons under AI substitution.*
- **Gap:** decline (del Rio-Chanona), composition/exit (Burtch), content (Helic) all done; **cohort conversion rate is not.**
- **vs Burtch:** *he measures inflow/exit (cross-sectional, ends Mar 2023); I measure the conversion rate of the intake (unconditional, folding attrition in). Different in kind.*
- **Methodology (Figure 1 seed):**

Join-cohorts (by quarter): PRE 2019-21 | POST 2023-24

| fixed 12-month window (constant)



Primary outcome (UNCONDITIONAL on persistence):

share of the join-cohort reaching first accepted answer within window



Identification: EVENT-STUDY DiD - HIGH vs LOW AI-substitutable tags

(pre-period flat = parallel-trends check; MUST PASS → else descriptive MVP)

unit = tag · cluster = tag · control = entry-activity threshold + cohort size

split-sample: develop → CONFIRM

- **Feedback I need:** (1) Is the unconditional conversion-rate framing distinct enough from Burtch? (2) Is the tag-level event-study credible here; is my *contemporaneous* substitutability proxy sound? (3) Where is your line on causal language for this design? (4) Is the entry-activity restriction enough for entry-composition — and do you agree the *unconditional* estimand avoids the persistence-collider problem? (5) One site enough? (6) If parallel trends fail, is the descriptive MVP acceptable to you?

Weekly roadmap

Week	Core	Parallel	Milestone	Gate
1	Ph.2 matrix → Ph.3 gap+contribution → Ph.4 lock causal core → Ph.5 feasibility + power + split → Ph.6 build+begin treatment → Ph.7 meeting	Skeleton + rough Intro; Decision Log live	Matrix; power verdict ; split declared; instructor go	A, B, C (T begun)
2	Finish Ph.6 (GATE T) → Ph.8 freeze + PAP → Ph.9 pipeline (snapshot, pinned env)	Draft Methods	Validated treatment; PAP committed; clean panel; async instructor check	T
3	Ph.10 exploratory (develop) → Ph.11 parallel-trends FIRST (GATE D) → confirmatory (confirm) + robustness	Draft Results; event-study figure	Identification verdict (+ MVP if failed); main estimate	D
4	Ph.12 interpret → lit re-scan → Ph.13 consolidate → Ph.14	Reference audit; sample+access+snapshot	Full draft → verified refs →	—

Week	Core	Parallel	Milestone	Gate
	audit + package → submit		repro zip → submit	
Buffer	~2 floating days across Wks 2-4	—	absorbs pipeline overrun	—

Immediate next actions (today)

1. **Lock the causal core (Phase 4):** unconditional estimand; primary outcome = share reaching first accepted answer within window; **unit = tag**; treatment-assignment rule. Log each in Track D.
2. **Set up Tracks R and D;** start the literature matrix (Phase 2).
3. **Write the contribution statement** and the paper skeleton + rough Intro (Track W).
4. **Run the Phase-5 feasibility check the v3 way:** computability + joins + censoring **and a hard count of post-cohort $N \times$ base conversion rate (power)** — no real pre/post peeking; then declare and seal the confirm set.
5. **Pre-choose the fallback site;** email the instructor; start the brief.

Highest-value action this week: the power/feasibility count. It is the cheapest way to learn — in Week 1 — whether the DiD is viable or whether you commit to the descriptive MVP with your eyes open.

Return each session: find your row, do the next task, log every real decision as you make it.